

Feature-Selection-Based Koopman Modeling and Robust LQR Control of Robots

First Author^{a,*}

^a*Institution Name, City, Country*

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ABSTRACT

Koopman operator theory with Extended Dynamic Mode Decomposition (EDMD) enables nonlinear dynamical systems to be represented by linear dynamics in a lifted space. However, the performance of EDMD strongly depends on the choice of basis functions: insufficient dictionaries limit approximation accuracy, whereas overcomplete dictionaries introduce redundancy, overfitting, and high computational cost. To address this issue, this paper proposes a feature-selection-based Koopman representation learning framework for robotic systems. A physics-informed overcomplete candidate library is first constructed, and a learnable feature selection matrix is then optimized to extract compact and informative linear combinations of candidate functions. The resulting modeling problem is formulated as a joint optimization over the Koopman matrices and the feature selection matrix, and a block coordinate descent algorithm is developed with convergence guarantees. Based on the learned lifted model, a feedforward-feedback Linear Quadratic Regulator controller is designed, where finite-dimensional approximation errors and external uncertainties are treated as bounded lumped disturbances. Lyapunov analysis proves that the closed-loop tracking error is ultimately uniformly bounded. Experiments on a three-degree-of-freedom robotic arm and a cable-driven soft platform demonstrate improved modeling accuracy, tracking performance, and robustness.

1. Introduction

Modern robotic systems such as manipulators, mobile robots, and soft robots exhibit highly nonlinear dynamics caused by mechanical coupling, actuator characteristics, and uncertain interactions with the environment [5, 18]. Accurate dynamic models are essential for high-performance control, but first-principles modeling becomes difficult when the robot contains flexible structures, uncertain parameters, or unmodeled interactions [14]. Consequently, data-driven models learned from operational data have become an important tool for robotic prediction and control [3, 21, 12, 11].

Koopman operator theory offers a control-oriented way to represent nonlinear dynamics by linear evolution in a lifted observable space [7, 9]. Since the Koopman operator is infinite-dimensional, practical methods rely on finite-dimensional approximations. Extended Dynamic Mode Decomposition (EDMD) constructs such an approximation by lifting the state through a user-specified dictionary of basis functions [24, 9]. The resulting lifted linear models enable the use of linear-system tools, including Koopman-based model predictive control and Koopman-based linear quadratic regulation, for nonlinear robotic systems [16, 8, 4].

A key bottleneck of EDMD is the choice of basis functions. In principle, the dictionary should span a subspace that is approximately invariant under the Koopman operator. In practice, constructing such a dictionary requires substantial prior knowledge of the underlying dynamics [15]. This leads to a tradeoff: a small dictionary may miss important

nonlinear features, whereas an overcomplete dictionary increases computational cost, introduces redundancy and ill-conditioning, and may cause overfitting [14, 20].

Existing studies mainly address this issue through learned dictionaries or sparse selection. Deep learning methods can learn nonlinear coordinates from data [13, 19, 17], but the resulting features are often difficult to interpret and analyze in robotic control. Sparse regression methods select compact representations from large libraries [10, 25], but they may involve nonsmooth optimization and sensitivity to noise and hyperparameters [23]. For robotic systems, it is therefore desirable to retain the interpretability of physics-informed candidate functions while avoiding the computational burden of using the full overcomplete dictionary.

To this end, this paper proposes a Koopman representation learning framework with a learnable feature selection mechanism. An overcomplete physics-informed candidate library is first constructed, and an optimizable selection matrix is then learned to extract compact linear combinations of the candidate functions. Unlike PCA-based dimensionality reduction, which preserves data variance, the proposed selection matrix is optimized directly for lifted one-step prediction accuracy. The modeling problem is formulated as a joint optimization over the Koopman matrices and the feature selection matrix, and a block coordinate descent algorithm is developed with convergence guarantees. Based on the learned lifted model, a feedforward-feedback Koopman-LQR controller is designed for trajectory tracking. The finite-dimensional Koopman approximation error is further treated as a bounded disturbance, which allows the closed-loop tracking error to be analyzed through Lyapunov arguments.

The innovations and contributions of this paper are summarized as follows.

*Corresponding author

 first.author@example.com (F. Author)

ORCID(s):

- (1) A feature-selection-based Koopman representation learning framework is proposed for robotic systems. By constructing a physics-informed overcomplete candidate function library and embedding a learnable feature selection matrix into the lifted state, the proposed method extracts compact and informative feature combinations while preserving the interpretability of the candidate library.
- (2) The Koopman modeling task is formulated as a joint optimization problem over the Koopman matrices and the feature selection matrix. The resulting problem is solved by a block coordinate descent algorithm, where the matrix update subproblems are explicitly characterized and convergence to a stationary point is established under standard assumptions.
- (3) A robust feedforward-feedback Koopman-LQR controller is developed for robotic trajectory tracking. By modeling finite-dimensional Koopman approximation errors and external uncertainties as bounded lumped disturbances, Lyapunov analysis proves that the closed-loop tracking error is ultimately uniformly bounded and relates the disturbance level to the steady-state tracking bound.

The rest of this paper is organized as follows. Section 2 introduces the preliminaries of Koopman operator theory and formulates the problem considered in this paper. Section 3 presents the feature-selection-based Koopman representation learning framework and the corresponding block coordinate descent algorithm. Section 4 develops the Koopman-based feedforward-feedback LQR controller and analyzes the closed-loop robust tracking performance. Section 5 reports the experimental validation results on a real robotic platform. Finally, Section 6 concludes this paper.

Notation: Throughout this paper, unless otherwise stated, $\|\cdot\|$ denotes the Euclidean norm for vectors and the induced 2-norm for matrices, $\|\cdot\|_1$ denotes the ℓ_1 norm, and $\|\cdot\|_F$ denotes the Frobenius norm for matrices; $\mathbb{R}^n$ denotes the n -dimensional Euclidean space; the superscripts $(\cdot)^\top$ and $(\cdot)^\dagger$ denote the transpose and the Moore–Penrose pseudoinverse, respectively; $\mathbf{I}_n$ denotes the $n \times n$ identity matrix; $\mathbf{0}_{m \times n}$ denotes the $m \times n$ zero matrix; $\text{tr}(\cdot)$, $\text{rank}(\cdot)$, and $\text{diag}(\cdot)$ denote the trace, rank, and diagonal matrix operator, respectively; $\mathcal{K}$ denotes the Koopman operator; $\mathcal{H}$ denotes a Hilbert space; $\Phi(\mathbf{x})$ denotes the feature library; for a matrix $\mathbf{A}$, $\mathbf{A} \geq 0$ and $\mathbf{A} > 0$ mean that $\mathbf{A}$ is positive semidefinite and positive definite, respectively.

2. Preliminaries and problem formulation

2.1. Koopman Operator Theory

Consider the discrete-time nonlinear system

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k), \quad (1)$$

where $\mathbf{x}_k \in \mathbb{R}^n$ is the system state and $\mathbf{u}_k \in \mathbb{R}^m$ is the control input at time step k . The mapping $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^n$ describes the state transition. It is assumed that f is Lipschitz continuous on its domain.

Let $\mathcal{H}$ be a Hilbert space of real-valued observation functions defined on the state space. For any observation function $g \in \mathcal{H}$, when the control input is $\mathbf{u}_k$, the discrete-time Koopman operator $\mathcal{K} : \mathcal{H} \rightarrow \mathcal{H}$ is defined by the evolution of g along the system trajectories as

$$(\mathcal{K}g)(\mathbf{x}_k, \mathbf{u}_k) \triangleq g(f(\mathbf{x}_k, \mathbf{u}_k)) = g(\mathbf{x}_{k+1}). \quad (2)$$

The core of Koopman operator theory is to lift the nonlinear dynamics from the finite-dimensional state space to an infinite-dimensional function space $\mathcal{H}$, thereby achieving linear evolution that obeys the superposition principle:

$$\mathcal{K}(\alpha g_1 + \beta g_2) = \alpha \mathcal{K}g_1 + \beta \mathcal{K}g_2, \quad (3)$$

where $g_1, g_2 \in \mathcal{H}$ are arbitrary observation functions and $\alpha, \beta \in \mathbb{R}$ are arbitrary scalars. This linearity enables the application of linear-system tools to analyze nonlinear dynamics after lifting.

2.2. Problem Formulation

As $\mathcal{H}$ is typically infinite-dimensional, EDMD approximates the Koopman operator in a finite-dimensional space spanned by a candidate function library $\{g_i\}_{i=1}^M$. The approximation quality depends critically on this library. A small library may miss essential nonlinear features, whereas an overcomplete library increases computational cost and may introduce redundancy and overfitting.

This accuracy–complexity tradeoff constitutes a central challenge in basis function selection for EDMD and related data-driven Koopman methods. To address this challenge and enable practical robotic control, this paper aims to: (1) learn a compact Koopman lifted representation from a physics-informed overcomplete candidate function library by using a feature selection matrix; and (2) design a robust feedforward-feedback LQR controller based on the learned lifted model to achieve reliable trajectory tracking under bounded lumped disturbances.

3. Koopman Representation Learning Based on Feature Selection Mechanism

This section develops the feature-selection-based Koopman learning framework, including the joint optimization algorithm and the lumped-disturbance model used for robust control.

3.1. Learnable Feature Selection and Joint Optimization

We first construct an overcomplete candidate observation function library $\Phi(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}^P$, where $P \gg n$, to provide sufficient nonlinear expressiveness. To remove redundant components while preserving the physics-informed structure of the library, a learnable feature selection matrix $\mathbf{D} \in \mathbb{R}^{N \times P}$ with $N \ll P$ is introduced. The resulting lifted coordinate is formally defined as follows.

Definition 3.1 (Feature-selected augmented state). *Given the candidate library $\Phi(\mathbf{x}_k)$ and the selection matrix $\mathbf{D}$, the*

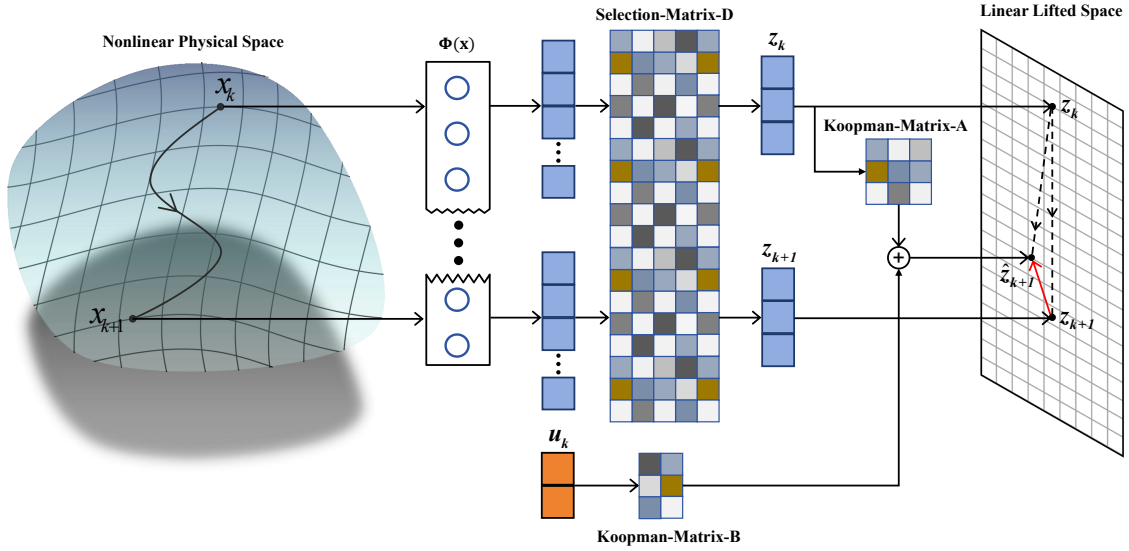


Figure 1: Koopman representation learning with the learnable feature selection mechanism.

compact feature vector and the augmented state are defined as

$$\eta_k = \mathbf{D}\Phi(\mathbf{x}_k), \quad \mathbf{z}_k = [\mathbf{x}_k^\top, \eta_k^\top]^\top \in \mathbb{R}^{n+N}. \quad (4)$$

Here, η_k preserves selected combinations of the overcomplete candidate functions, while the original physical state remains explicitly embedded in $\mathbf{z}_k$.

Based on this augmented representation, we seek the following linear evolution model in the lifted space:

$$\mathbf{z}_{k+1} = \mathbf{A}\mathbf{z}_k + \mathbf{B}\mathbf{u}_k, \quad (5)$$

where $\mathbf{A} \in \mathbb{R}^{(n+N) \times (n+N)}$ is the Koopman matrix governing the linear dynamics in the lifted space, $\mathbf{B} \in \mathbb{R}^{(n+N) \times m}$ is the input matrix relating the control input to the augmented state evolution, and $\mathbf{K} \in \mathbb{R}^{(n+N) \times (n+N+m)}$ is the augmented Koopman matrix defined as $\mathbf{K} \triangleq [\mathbf{A}, \mathbf{B}]$. This structure captures the nonlinear dynamics of the system through $\mathbf{D}\Phi$, while simultaneously allowing lossless recovery of the physical state via the projection matrix $\mathbf{C} = [\mathbf{I}_n \ \mathbf{0}_{n \times N}]$ according to

$$\mathbf{x}_k = \mathbf{C}\mathbf{z}_k, \quad (6)$$

which facilitates the subsequent full-state feedback design. Figure 1 illustrates the proposed Koopman representation learning framework.

Given a trajectory dataset $\{\mathbf{x}_k, \mathbf{u}_k\}_{k=1}^{T+1}$, where $\mathbf{x}_k \in \mathbb{R}^n$ represents the robot state (e.g., joint positions and velocities) and $\mathbf{u}_k \in \mathbb{R}^m$ denotes the control torque applied to each joint, which yields T one-step transition pairs, we construct the state snapshot matrices $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_T]$ and $\mathbf{Y} = [\mathbf{x}_2, \dots, \mathbf{x}_{T+1}]$, the input matrix $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_T]$, and the feature matrices $\mathbf{X}_\Phi = [\Phi(\mathbf{x}_1), \dots, \Phi(\mathbf{x}_T)]$ and $\mathbf{Y}_\Phi = [\Phi(\mathbf{x}_2), \dots, \Phi(\mathbf{x}_{T+1})]$. For a given $\mathbf{D}$, the lifted state data matrices are defined as

$$\mathbf{Z}(\mathbf{D}) = \begin{bmatrix} \mathbf{X} \\ \mathbf{D}\mathbf{X}_\Phi \end{bmatrix}, \quad \mathbf{Z}^+(\mathbf{D}) = \begin{bmatrix} \mathbf{Y} \\ \mathbf{D}\mathbf{Y}_\Phi \end{bmatrix}. \quad (7)$$

We formulate the joint optimization problem as

$$\begin{aligned} L(\mathbf{K}, \mathbf{D}) &= \frac{1}{2} \left\| \mathbf{Z}^+(\mathbf{D}) - \mathbf{K} \begin{bmatrix} \mathbf{Z}(\mathbf{D}) \\ \mathbf{U} \end{bmatrix} \right\|_F^2 \\ &\quad + \lambda \|\mathbf{D}\|_1 + \beta \|\mathbf{K}\|_F^2, \\ (\mathbf{K}^*, \mathbf{D}^*) &= \arg \min_{\mathbf{K}, \mathbf{D}} L(\mathbf{K}, \mathbf{D}). \end{aligned} \quad (8)$$

where $\lambda > 0$ and $\beta > 0$ are the regularization coefficients. The ℓ_1 regularization on $\mathbf{D}$ promotes sparse linear combinations of informative candidate functions from the overcomplete candidate function library. Meanwhile, the objective function enforces accurate one-step prediction of both the physical state and the projected feature components in the lifted state, thereby discouraging trivial solutions for $\mathbf{D}$. When $\mathbf{D} = \mathbf{I}_P$ and the regularization terms $\lambda \|\mathbf{D}\|_1$ and $\beta \|\mathbf{K}\|_F^2$ are removed, the formulation reduces to standard EDMD.

Since the optimization problem in (8) is nonconvex in the joint variables $(\mathbf{K}, \mathbf{D})$, we adopt a two-block coordinate descent (BCD) algorithm [22] to solve it by alternately updating the two variable blocks. Let $(\mathbf{K}^{(t)}, \mathbf{D}^{(t)})$ denote the iterates at iteration t , and define the corresponding objective value as

$$L^{(t)} := L(\mathbf{K}^{(t)}, \mathbf{D}^{(t)}).$$

At each iteration, the two variable blocks are updated alternately by solving a strongly convex $\mathbf{K}$ -subproblem and a convex $\mathbf{D}$ -subproblem.

1) **K-subproblem:** With $\mathbf{D}^{(t)}$ fixed, define

$$\mathbf{Y}_{\text{aug}}^{(t)} := \mathbf{Z}^+(\mathbf{D}^{(t)}), \quad \mathbf{W}^{(t)} := \begin{bmatrix} \mathbf{Z}(\mathbf{D}^{(t)}) \\ \mathbf{U} \end{bmatrix}.$$

Then, the $\mathbf{K}$ -subproblem is given by

$$\min_{\mathbf{K}} \frac{1}{2} \|\mathbf{Y}_{\text{aug}}^{(t)} - \mathbf{K}\mathbf{W}^{(t)}\|_F^2 + \beta \|\mathbf{K}\|_F^2.$$

This is a standard ridge regression problem. Setting the gradient with respect to $\mathbf{K}$ to zero gives

$$-(\mathbf{Y}_{\text{aug}}^{(t)} - \mathbf{K}\mathbf{W}^{(t)})\mathbf{W}^{(t)\top} + 2\beta\mathbf{K} = \mathbf{0},$$

from which the closed-form update is derived as

$$\mathbf{K}^{(t+1)} = \mathbf{Z}^+(\mathbf{D}^{(t)})\mathbf{W}^{(t)\top} (\mathbf{W}^{(t)}\mathbf{W}^{(t)\top} + 2\beta\mathbf{I})^{-1}. \quad (9)$$

For the $\mathbf{K}$ update, the ridge regularization makes the subproblem strongly convex for any fixed $\mathbf{D}^{(t)}$ and $\beta > 0$, so the minimizer in (9) is unique.

2) **D-subproblem:** With $\mathbf{K}^{(t+1)}$ fixed, the $\mathbf{D}$ -subproblem is formulated as

$$\mathbf{D}^{(t+1)} = \arg \min_{\mathbf{D}} J^{(t)}(\mathbf{D}), \quad (10)$$

where

$$J^{(t)}(\mathbf{D}) = \frac{1}{2} \left\| \mathbf{Z}^+(\mathbf{D}) - \mathbf{K}^{(t+1)} \begin{bmatrix} \mathbf{Z}(\mathbf{D}) \\ \mathbf{U} \end{bmatrix} \right\|_F^2 + \lambda \|\mathbf{D}\|_1. \quad (11)$$

To see this, we write $\mathbf{K}^{(t+1)} = [\mathbf{A}^{(t+1)}, \mathbf{B}^{(t+1)}]$ and partition $\mathbf{A}^{(t+1)}$ according to the augmented state structure. Since both $\mathbf{Z}(\mathbf{D})$ and $\mathbf{Z}^+(\mathbf{D})$ depend affinely on $\mathbf{D}$, the residual

$$\mathbf{Z}^+(\mathbf{D}) - \mathbf{K}^{(t+1)} \begin{bmatrix} \mathbf{Z}(\mathbf{D}) \\ \mathbf{U} \end{bmatrix}$$

is affine in $\mathbf{D}$. Hence, the data-fitting term in (11) is a convex quadratic function of $\mathbf{D}$, and the subproblem can be efficiently solved using proximal gradient methods [1].

For the $\mathbf{D}$ update, the subproblem in (10) is a convex composite optimization problem consisting of a quadratic data-fitting term and a separable ℓ_1 regularizer. Hence, it can be solved by standard proximal gradient updates.

Let $T_{\max}$ denote the maximum number of iterations and let $\varepsilon > 0$ denote the convergence tolerance. The above block structure leads to the BCD procedure summarized in Algorithm 1. Its convergence property is stated as follows.

Theorem 3.1 (Convergence of the BCD Algorithm). *Suppose that $\lambda > 0$ and $\beta > 0$, and that each subproblem is solved to optimality at every iteration. Then the sequence $\{(\mathbf{K}^{(t)}, \mathbf{D}^{(t)})\}$ generated by Algorithm 1 satisfies:*

1. *The objective values $L^{(t)}$ are monotonically nonincreasing and converge to a finite limit;*
2. *The iterates $\{(\mathbf{K}^{(t)}, \mathbf{D}^{(t)})\}$ remain bounded;*
3. *Every accumulation point of the sequence is a first-order stationary point of (8).*

Proof. First, since each block is minimized exactly while the other block is fixed, the two successive updates in Algorithm 1 yield

$$\begin{aligned} L(\mathbf{K}^{(t+1)}, \mathbf{D}^{(t)}) &\leq L(\mathbf{K}^{(t)}, \mathbf{D}^{(t)}), \\ L(\mathbf{K}^{(t+1)}, \mathbf{D}^{(t+1)}) &\leq L(\mathbf{K}^{(t+1)}, \mathbf{D}^{(t)}). \end{aligned}$$

Therefore, $L^{(t+1)} \leq L^{(t)}$, and the objective sequence is monotonically nonincreasing. Since $L(\mathbf{K}, \mathbf{D}) \geq 0$, it converges to a finite limit.

Second, from $L^{(t)} \leq L^{(0)}$ and the nonnegativity of the data-fitting term, we obtain

$$\beta \|\mathbf{K}^{(t)}\|_F^2 + \lambda \|\mathbf{D}^{(t)}\|_1 \leq L^{(0)}.$$

Hence, the sequence $\{\mathbf{K}^{(t)}\}$ is bounded. The sequence $\{\mathbf{D}^{(t)}\}$ is also bounded, because boundedness in the ℓ_1 norm implies boundedness in the Frobenius norm for finite-dimensional matrices.

Finally, the objective in (8) is the sum of a continuously differentiable data-fitting term and a convex separable nonsmooth regularization term, and each block subproblem is solved to optimality at every iteration. By the standard convergence results for block coordinate descent applied to nonsmooth objectives [22], every accumulation point of $\{(\mathbf{K}^{(t)}, \mathbf{D}^{(t)})\}$ is a first-order stationary point of (8). $\square$

Algorithm 1 Block Coordinate Descent Algorithm for Koopman Representation Learning

Input: Data $(\mathbf{X}, \mathbf{Y}, \mathbf{U})$, basis library Φ , parameters λ, β , initialization $\mathbf{D}^{(0)}$

Output: Learned $\mathbf{K}^*$ and $\mathbf{D}^*$

- 1: **for** $t = 0, 1, \dots, T_{\max} - 1$ **do**
 - 2: Update $\mathbf{K}^{(t+1)}$ using (9)
 - 3: Update $\mathbf{D}^{(t+1)}$ by solving (10) via a proximal gradient method
 - 4: **if** $|L^{(t+1)} - L^{(t)}| < \varepsilon$ **then**
 - 5: **break**
 - 6: **end if**
 - 7: **end for**
-

3.2. Lumped Disturbance Analysis

The learned Koopman model captures the dominant system dynamics, but the finite-dimensional lifted model inevitably contains approximation residuals. In addition, sensor noise and environmental disturbances may appear during robot operation. We collect these effects into a lumped disturbance term $\mathbf{w}_k$, and the actual evolution in the lifted space is written as

$$\mathbf{z}_{k+1} = \mathbf{A}\mathbf{z}_k + \mathbf{B}\mathbf{u}_k + \mathbf{w}_k, \quad (12)$$

where $\mathbf{A}$ and $\mathbf{B}$ are the learned Koopman state-transition matrix and input matrix, respectively, obtained from Algorithm 1.

When the state and input are constrained to compact sets, the Koopman approximation error is theoretically bounded [8]. Sensor noise and environmental disturbances are also bounded in practical robotic systems due to hardware limits and operating conditions. This motivates the following assumption.

Assumption 3.1 (Bounded lumped disturbance). *There exists a constant $\delta_w > 0$ such that the lumped disturbance in (12) satisfies*

$$\|\mathbf{w}_k\| \leq \delta_w, \quad \forall k. \quad (13)$$

Assumption 3.1 provides the interface between the learned finite-dimensional Koopman model and the robust tracking analysis. A smaller approximation residual corresponds to a smaller admissible disturbance level, which is reflected in the ultimate tracking bound derived in Section 4.2.

4. LQR Control Based on Koopman Model

This section designs a feedforward-feedback Koopman-LQR controller for the disturbed lifted model (12) and establishes the ultimate boundedness of the closed-loop tracking error.

4.1. Feedforward-Feedback Robust Controller Design

To achieve robust trajectory tracking, we design the controller based on the disturbed augmented linear model (12) obtained from the learned Koopman model in Section 3. First, let $\{\mathbf{x}_k^{\text{ref}}\}_{k=0}^{\infty}$ denote the desired reference trajectory in the physical state space. By lifting this trajectory through the feature mapping and the learned selection matrix $\mathbf{D}^*$, we obtain the corresponding lifted reference trajectory:

$$\mathbf{z}_k^{\text{ref}} = [\mathbf{x}_k^{\text{ref}^\top}, (\mathbf{D}^* \Phi(\mathbf{x}_k^{\text{ref}}))^\top]^\top. \quad (14)$$

To balance the tracking accuracy and the control efforts, we define the following infinite-horizon quadratic cost function:

$$J = \sum_{k=0}^{\infty} ((\mathbf{z}_k - \mathbf{z}_k^{\text{ref}})^\top \mathbf{Q} (\mathbf{z}_k - \mathbf{z}_k^{\text{ref}}) + \mathbf{u}_k^\top \mathbf{R} \mathbf{u}_k), \quad (15)$$

where $\mathbf{Q} = \mathbf{C}^\top \mathbf{Q}_x \mathbf{C} \geq 0$ is the lifted-state weight matrix, $\mathbf{Q}_x$ is the weight matrix in the original state space, and $\mathbf{R} > 0$ is the control input weight matrix.

Assume that the physical system is controllable and that the training data are sufficiently informative such that the learned linear model $(\mathbf{A}, \mathbf{B})$ is stabilizable. Under these conditions, the standard LQR theory yields a unique optimal state-feedback gain $\mathbf{K}_{\text{lqr}}$, which renders the closed-loop matrix $\mathbf{A}_{\text{cl}} = \mathbf{A} - \mathbf{B}\mathbf{K}_{\text{lqr}}$ Schur stable.

To guarantee the trajectory tracking of the robots, we propose the following composite feedforward-feedback control law:

$$\mathbf{u}_k = \mathbf{u}_k^{\text{ff}} - \mathbf{K}_{\text{lqr}}(\mathbf{z}_k - \mathbf{z}_k^{\text{ref}}). \quad (16)$$

The feedforward term $\mathbf{u}_k^{\text{ff}}$ is used to compensate for the known dynamics along the reference trajectory and is computed as

$$\mathbf{u}_k^{\text{ff}} = \mathbf{B}^\dagger(\mathbf{z}_{k+1}^{\text{ref}} - \mathbf{A}\mathbf{z}_k^{\text{ref}}), \quad (17)$$

where $\mathbf{B}^\dagger$ denotes the Moore-Penrose pseudoinverse of $\mathbf{B}$. Figure 2 illustrates the structure of the proposed Koopman-LQR controller.

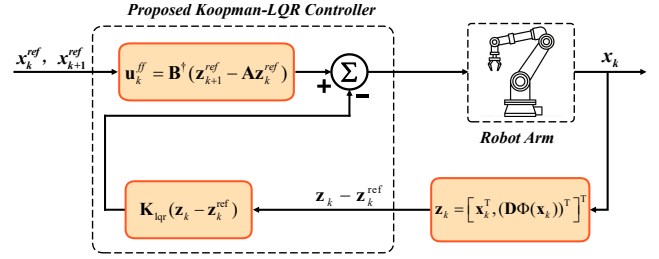


Figure 2: The proposed Koopman-based feedforward-feedback LQR control structure.

4.2. Stability Analysis

Based on the feedforward-feedback controller (16), we present the following stability result.

The key observation is that the feedforward term removes the nominal reference dynamics, whereas the LQR feedback stabilizes the deviation around the lifted reference trajectory. Therefore, the closed-loop tracking error can be analyzed as a stable linear system driven by the lumped disturbance.

Theorem 4.1 (Robust Tracking Performance). *Consider the system (12) under the control law (16). Suppose that Assumption 3.1 holds. Then the closed-loop system achieves ultimately uniformly bounded (UUB) tracking, i.e., there exists $R > 0$ such that, for any initial condition, there exists a finite time K for which $\|\mathbf{e}_k\| \leq R$ for all $k \geq K$.*

Proof. Define the tracking error as $\mathbf{e}_k := \mathbf{z}_k - \mathbf{z}_k^{\text{ref}}$. Substituting the control law (16) into the disturbed augmented model (12), and using the definition of the feedforward term in (17), the closed-loop error dynamics are given by

$$\mathbf{e}_{k+1} = (\mathbf{A} - \mathbf{B}\mathbf{K}_{\text{lqr}})\mathbf{e}_k + \mathbf{w}_k = \mathbf{A}_{\text{cl}}\mathbf{e}_k + \mathbf{w}_k, \quad (18)$$

where $\mathbf{A}_{\text{cl}} := \mathbf{A} - \mathbf{B}\mathbf{K}_{\text{lqr}}$ denotes the closed-loop system matrix. By the stabilizability assumption in Section 4.1 and the standard LQR theory, $\mathbf{A}_{\text{cl}}$ is Schur stable. Hence, there exist positive definite matrices $\mathbf{P} > 0$ and $\mathbf{S} > 0$ satisfying the discrete-time Lyapunov equation

$$\mathbf{A}_{\text{cl}}^\top \mathbf{P} \mathbf{A}_{\text{cl}} - \mathbf{P} = -\mathbf{S}.$$

Consider the Lyapunov function $V_k = \mathbf{e}_k^\top \mathbf{P} \mathbf{e}_k$. Then, along the trajectories of (18),

$$\begin{aligned} \Delta V_k &= V_{k+1} - V_k \\ &= \mathbf{e}_k^\top (\mathbf{A}_{\text{cl}}^\top \mathbf{P} \mathbf{A}_{\text{cl}} - \mathbf{P}) \mathbf{e}_k + 2\mathbf{e}_k^\top \mathbf{A}_{\text{cl}}^\top \mathbf{P} \mathbf{w}_k + \mathbf{w}_k^\top \mathbf{P} \mathbf{w}_k \\ &= -\mathbf{e}_k^\top \mathbf{S} \mathbf{e}_k + 2\mathbf{e}_k^\top \mathbf{A}_{\text{cl}}^\top \mathbf{P} \mathbf{w}_k + \mathbf{w}_k^\top \mathbf{P} \mathbf{w}_k, \end{aligned} \quad (19)$$

where the last equality follows from the discrete-time Lyapunov equation. By the boundedness of $\mathbf{w}_k$ and the Cauchy-Schwarz inequality, it follows that when $\|\mathbf{e}_k\|$ exceeds a threshold R determined by δ_w , one has $\Delta V_k < 0$. Therefore, the error trajectory enters and remains within the bounded set $\mathcal{B}_R = \{\mathbf{e} : \|\mathbf{e}\| \leq R\}$ [2, 6]. $\square$

Since the projection matrix $\mathbf{C}$ satisfies $\|\mathbf{C}\| = 1$, the error bound in the lifted space directly guarantees the tracking accuracy in the original state space:

$$\|\mathbf{x}_k - \mathbf{x}_k^{\text{ref}}\| = \|\mathbf{C}\mathbf{e}_k\| \leq \|\mathbf{e}_k\| \leq R.$$

Furthermore, the steady-state error bound can be upper bounded by a linear function of the disturbance level, i.e.,

$$R \leq c \delta_w,$$

where the constant $c > 0$ depends only on the spectral properties of the closed-loop matrix $\mathbf{A}_{\text{cl}}$ [2]. This relationship connects the Koopman modeling accuracy with the closed-loop performance: reducing the approximation component of $\mathbf{w}_k$ decreases δ_w and hence shrinks the ultimate tracking bound.

5. Experimental Validations

5.1. Experimental Platform and Data Collection

Experiments were conducted on a Geomagic Touch[®] robot arm, whose mechanical structure is shown in Fig. 3. The device has three revolute joints with integrated encoders for real-time measurement of joint position and velocity. The force-feedback stylus at the end-effector was kept fixed during the experiments so that the joint-space dynamics could be isolated.

According to the technical specifications provided by the manufacturer, the motion ranges of joints 1, 2, and 3 are $[-1.3, 0.66]$ rad, $[0.2, 2.0]$ rad, and $[-0.65, 0.87]$ rad, respectively. Joints 2 and 3 are mechanically coupled, so the allowable range of joint 3 is affected when joint 2 approaches its motion limit. To reduce the influence of this coupling, the data collection experiments were conducted to avoid the coupled boundary as much as possible.

During the data collection experiments, continuous random control torques were applied to the three joints to excite the system dynamics. The sampling period was 0.004 s. In total, 20 trajectories were collected, each with a duration of 20 s, containing joint positions, velocities, and the corresponding control inputs. The complete dataset was split into training, validation, and test sets with a ratio of 15:4:1.

The second platform is a cable-driven soft robotic system. Its state is the two-dimensional light-spot position in the xy plane, and its two-dimensional input commands the cable-driven motion of the light spot. The dataset contains 100 trajectories with 300 samples each. A fixed by-trajectory 70/20/10 split is used, and trajectory 66 is used as the displayed modeling test case.

5.2. Model Prediction Accuracy Comparison

For comparison, two Koopman baselines were used: EDMDDL [10], which learns a 60-dimensional dictionary using a five-layer neural network with 512 ReLU neurons per layer, and standard EDMD [24], which directly uses the full 364-dimensional candidate function library. FS-EDMD uses the same candidate function library as standard EDMD but learns a compact selection matrix $\mathbf{D} \in \mathbb{R}^{60 \times 364}$. All



Figure 3: The Geomagic Touch robot arm.

methods were trained on the same training set. For the robotic arm, evaluation uses an 800-step continuous rollout on an independent test trajectory. For the soft platform, the three models are compared on test trajectory 66 over its full rollout.

Fig. 4 shows the prediction results. On the robotic arm, FS-EDMD maintains the closest agreement with the measured joint trajectory over the 3.2 s prediction horizon, whereas EDMDDL gradually drifts and standard EDMD accumulates larger errors. On the soft platform, the x and y predictions of FS-EDMD follow the measured light-spot trajectory more closely than the two baselines over the displayed test trajectory.

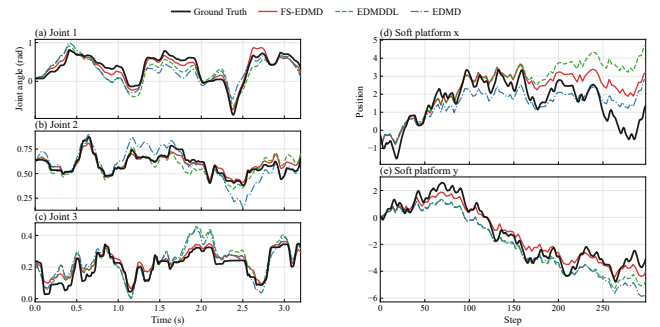


Figure 4: Modeling accuracy on the robotic arm and the cable-driven soft platform. The left panels show Joint 1–3 predictions over the 800-step rollout, and the right panels show the x and y predictions for soft-platform test trajectory 66.

As shown in Table 1, FS-EDMD achieves the lowest full-state prediction errors on the robotic arm, with MSE, RMSE, and MAE of 0.3520, 0.5933, and 0.3202, respectively.¹ These correspond to reductions of 40.7%, 23.0%, and 29.6% over EDMDDL, and 51.5%, 30.3%, and 36.2% over standard EDMD. On the soft platform, the FS-EDMD RMSE for test trajectory 66 is 0.7889, compared with 0.8965 for standard EDMD and 1.2927 for EDMDDL.

¹ $\text{MSE} = \frac{1}{T} \sum_{k=1}^T \|\mathbf{x}_k - \hat{\mathbf{x}}_k\|^2$, $\text{RMSE} = \sqrt{\text{MSE}}$, $\text{MAE} = \frac{1}{T} \sum_{k=1}^T \|\mathbf{x}_k - \hat{\mathbf{x}}_k\|$, where $\hat{\mathbf{x}}_k$ denotes the predicted state at time step k .

Table 1
Modeling Accuracy Comparison

Platform	Method	MSE	RMSE	MAE
Robotic arm	FS-EDMD	0.3520	0.5933	0.3202
	EDMDDL	0.5936	0.7704	0.4550
	EDMD	0.7256	0.8518	0.5022
Soft platform	FS-EDMD	—	0.7889	—
	EDMD	—	0.8965	—
	EDMDDL	—	1.2927	—

5.3. Trajectory Tracking Control Performance Evaluation

Trajectory tracking experiments were conducted in the joint space using sinusoidal and irregular references. The Koopman-LQR controller was implemented with the FS-EDMD model and the feedforward-feedback structure in Section 4. The weighting matrices were set to

$$\mathbf{Q} = \text{diag}(100, 10, 100, 10, 100, 10), \quad \mathbf{R} = \mathbf{I}_3,$$

and a PID controller with empirically tuned gains $K_p = 25$, $K_i = 0.1$, and $K_d = 2.5$ was used as the baseline. All tracking metrics were computed over the common 10–40 s evaluation window after controller initialization.

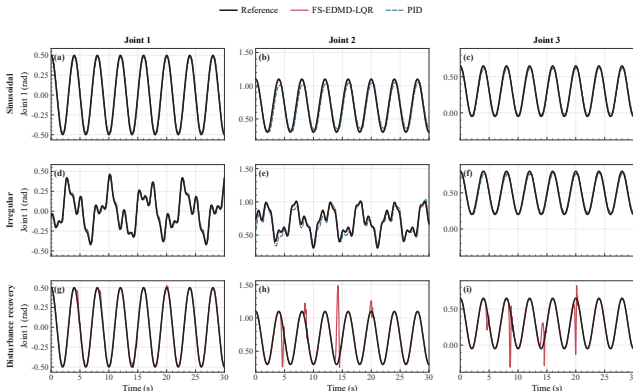


Figure 5: Robotic-arm trajectory tracking under sinusoidal, irregular, and disturbance-recovery references. Rows correspond to the three experiments and columns correspond to Joint 1–3. The first two rows compare Reference, FS-EDMD-LQR, and PID; the third row shows the disturbance-recovery response of FS-EDMD-LQR.

For sinusoidal tracking, the reference trajectories for the three joints were defined as

$$q_i^{\text{ref}}(t) = A_i \sin(\omega_i t) + b_i, \quad i = 1, 2, 3, \quad (20)$$

where q_i^{ref} denotes the desired joint position. The amplitudes were set to $A_1 = 0.5$ rad, $A_2 = 0.4$ rad, and $A_3 = 0.35$ rad, with $f_i = 0.25$ Hz and $b_i = 0$ for all joints. The

Robotic-arm metrics cover joint positions and velocities, whereas soft-platform metrics cover the xy light-spot position.

Table 2
Trajectory Tracking Performance

Type	Controller	RMSE	MAE
Arm: sinusoidal	FS-EDMD-LQR	0.0154	0.0136
	PID	0.0475	0.0402
Arm: irregular	FS-EDMD-LQR	0.0175	0.0148
	PID	0.0413	0.0337
Arm: disturbance recovery	FS-EDMD-LQR	0.0958	0.0298
	PID	0.6386	0.4797
Soft: figure-eight	FS-EDMD-LQR	0.1174	0.1135
	PID	0.6386	0.4797
Soft: five-point star	FS-EDMD-LQR	0.1339	0.1294
	PID	0.8406	0.6391

sinusoidal row of Fig. 5 shows that FS-EDMD-LQR tracks the references more accurately than PID, especially on joints 2 and 3.

For irregular tracking, joints 1 and 2 used composite references formed by the superposition of three sinusoidal components:

$$q_j^{\text{ref}}(t) = \sum_{l=1}^3 A_{j,l} \sin(2\pi f_{j,l} t + \phi_{j,l}), \quad j = 1, 2, \quad (21)$$

where $A_{j,l}$, $f_{j,l}$, and $\phi_{j,l}$ denote the amplitude, frequency, and phase of the l -th component, respectively. The component frequencies were (0.15, 0.40, 0.80) Hz for joint 1 and (0.18, 0.45, 0.90) Hz for joint 2, with amplitudes and phase shifts selected within the experimentally validated operating range. Due to the mechanical coupling between joints 2 and 3, joint 3 used $q_3^{\text{ref}}(t) = 0.35 \sin(2\pi \cdot 0.25t)$.

The irregular row of Fig. 5 shows that FS-EDMD-LQR follows the composite reference more accurately during fast transitions, whereas PID produces larger tracking errors.

Table 2 summarizes the quantitative tracking results.² FS-EDMD-LQR reduces the RMSE²/MAE by 67.5%/66.2% for sinusoidal tracking and by 57.7%/56.0% for irregular tracking, confirming its advantage over PID.

For the soft platform, both controllers used the same reference trajectory, initial condition, sampling period, input limit, and endpoint condition. To isolate trajectory tracking from the initial movement toward the reference start, RMSE and MAE were computed over the same samples after a common 30-step approach interval. Fig. 6 shows that FS-EDMD-LQR follows both the figure-eight and five-point-star references more closely than PID. As reported in Table 2, FS-EDMD-LQR reduces the RMSE²/MAE by 81.6%/76.3% on the figure-eight trajectory and by 84.1%/79.8% on the five-point-star trajectory.

²For the robotic arm, $\text{RMSE} = \sqrt{\frac{1}{3T} \sum_{i=1}^3 \sum_{k=1}^T (q_i(k) - q_i^{\text{ref}}(k))^2}$ and $\text{MAE} = \frac{1}{3T} \sum_{i=1}^3 \sum_{k=1}^T |q_i(k) - q_i^{\text{ref}}(k)|$. For the soft platform,

$\text{RMSE} = \sqrt{\frac{1}{T} \sum_{k=1}^T \|\mathbf{p}_k - \mathbf{p}_k^{\text{ref}}\|_2^2}$ and $\text{MAE} = \frac{1}{T} \sum_{k=1}^T \|\mathbf{p}_k - \mathbf{p}_k^{\text{ref}}\|_2$. Robotic-arm errors are expressed in rad, whereas soft-platform errors use the recorded light-spot position coordinates.

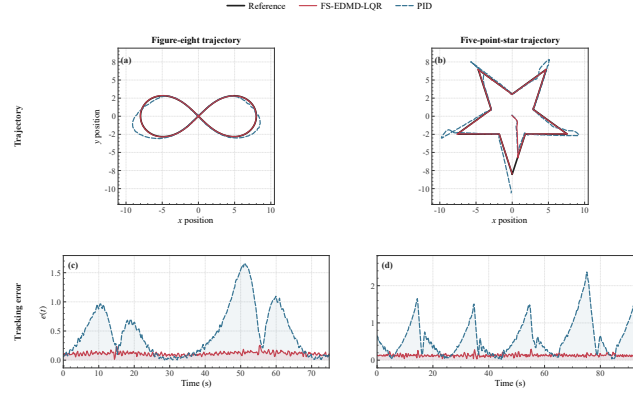


Figure 6: Soft-platform trajectory tracking and instantaneous tracking error on the figure-eight and five-point-star references. The top row compares the complete Reference, FS-EDMD-LQR, and PID trajectories; the bottom row shows the pointwise Euclidean tracking error after the common initial approach interval.

5.4. Robustness Validation Experiments

Robustness was evaluated using the sinusoidal tracking task with the same controller parameters. During the 30 s experiment, an operator manually disturbed the robot by briefly pushing and pulling the links associated with the three joints, thereby introducing unknown bounded joint torques. In the disturbance-recovery row of Fig. 5, the joint trajectories exhibit temporary deviations from the references during the disturbances but recover smoothly after the disturbances are removed and remain bounded throughout the experiment.

6. Conclusion

In this paper, we have presented a Koopman representation learning framework with a learnable feature selection matrix to address the tradeoff among modeling accuracy, computational efficiency, and interpretability in dictionary design for EDMD. By constructing an overcomplete physics-informed candidate function library and learning compact feature combinations through an optimizable selection matrix, the proposed method achieves accurate and interpretable Koopman modeling for robotic systems. Based on the learned model, a feedforward–feedback Koopman-LQR controller has been developed, and the ultimate boundedness of the closed-loop tracking error has been established under bounded disturbances. Experimental results on a three-degree-of-freedom robotic arm and a cable-driven soft platform demonstrate improved prediction accuracy, trajectory tracking performance, and robustness against external disturbances. Future work will consider extending the proposed framework to online learning settings and incorporating model predictive control to explicitly account for state and input constraints.

CRedit authorship contribution statement

First Author: Conceptualization, Methodology, Software, Writing – original draft.

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